

Finite Linear-Group Permutation Actions

Exact σ -MGFs, stable coefficients, and matrix verification

Proof and computational verification companion

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Abstract

For a finite group acting on a set S , we prove the determinant/cycle formula, its fixed-point reconstruction, and its interpretation as an orbit count on tuples of multisets. We then specialize to nonzero-vector, projective-point, and complete-flag actions of finite linear groups. Exact finite-field matrix enumerations verify all coefficients through degree three for $PSL_3(2)$, the degree-two vector claims for $GL_2(3)$, and the Bruhat obstruction in ranks two and three. The computation also exposes a sharp boundary phenomenon for $SL_2(3)$.

1 The universal finite permutation formula

Let G act on a finite set S , let $V = \mathbb{C}[S]$, and write P_g for the associated permutation matrix. For a partition $\tau = (\tau_1, \dots, \tau_s)$ set

$$W_\tau = \bigotimes_{b=1}^s \text{Sym}^{\tau_b}(V).$$

The symmetric-power identity

$$\det(I - tP_g)^{-1} = \sum_{r \geq 0} \text{tr}(P_g \mid \text{Sym}^r V) t^r$$

implies, after averaging over G ,

$$[m_\tau] \mathcal{S}_{G,S} = \frac{1}{|G|} \sum_{g \in G} \text{tr}(g \mid W_\tau) = \dim W_\tau^G. \quad (1)$$

The last equality follows because $|G|^{-1} \sum_{g \in G} g$ is the Reynolds projector.

If g has $c_d(g; S)$ cycles of length d , a single d -cycle has all d th roots of unity as eigenvalues and hence determinant $1 - t^d$. Therefore

$$\mathcal{S}_{G,S}(t_\bullet) = \frac{1}{|G|} \sum_{g \in G} \prod_{i \geq 1} \prod_{d \geq 1} (1 - t_i^d)^{-c_d(g; S)}. \quad (2)$$

This is an exact finite formula, not an asymptotic statement.

Fixed points determine the cycles

A point on a d -cycle is fixed by g^r exactly when $d \mid r$. Thus

$$\text{Fix}_S(g^r) = \sum_{d \mid r} d c_d(g; S).$$

Möbius inversion gives

$$c_d(g; S) = \frac{1}{d} \sum_{r \mid d} \mu(d/r) \text{Fix}_S(g^r). \quad (3)$$

For $S = G/P$, $\text{Fix}_S(g^r)$ is the value at g^r of the permutation character $\text{Ind}_P^G \mathbf{1}$. Grouping (2) by conjugacy class then yields the stated centralizer-weighted formula.

Direct orbit interpretation

A monomial basis vector of $\text{Sym}^r \mathbb{C}[S]$ is a multiset of r points of S . Consequently a basis of W_τ is a tuple of such multisets, of respective sizes $\tau_1, \dots, \tau_s$. Since G permutes this basis, the sum of the basis vectors in each orbit is one invariant basis vector. Hence

$$[m_\tau] \mathcal{S}_{G,S} = |\{G\text{-orbits on tuples of multisets of sizes } \tau_1, \dots, \tau_s\}|. \quad (4)$$

Equation (4) supplies the independent direct route used in the computation.

2 Linear and projective specializations

For $S = \mathbb{F}_q^n \setminus \{0\}$ and $g \in GL_n(q)$,

$$\text{Fix}_S(g^r) = q^{\dim \ker(g^r - I)} - 1. \quad (5)$$

For projective points, a fixed line is an eigenline whose eigenvalue lies in $\mathbb{F}_q^\times$. The eigenspaces for distinct eigenvalues contribute disjoint sets of lines, so

$$\text{Fix}_{\mathbb{P}^{n-1}(\mathbb{F}_q)}(g^r) = \sum_{\lambda \in \mathbb{F}_q^\times} \frac{q^{\dim \ker(g^r - \lambda I)} - 1}{q - 1}. \quad (6)$$

Substitution of (5) or (6) into (3) and then (2) proves the proposed exact formulas.

Over $\mathbb{F}_2$, every projective line has one nonzero vector and $GL_n(2) = SL_n(2) = PGL_n(2) = PSL_n(2)$. Thus (5) directly gives the displayed $PSL_n(2)$ formula.

Stable range

A configuration of total degree $D = |\tau|$ involves at most D projective points and its span has dimension at most D . When $n \geq D$, two such configurations are $GL_n(q)$ -equivalent exactly when the induced colored configurations on their spans are linearly isomorphic: any isomorphism of the spans extends after choosing complements. This proves coefficientwise stability for GL/PGL projective actions. The same argument for nonzero vectors is immediate. For SL/PSL , the determinant of an extension can be corrected on an unused one-dimensional summand, which requires the strict range $n > D$.

For projective points the stable expansion through degree three is

$$1 + m_{(1)} + 2m_{(2)} + 2m_{(1,1)} + 4m_{(3)} + 5m_{(2,1)} + 6m_{(1,1,1)} + O(\deg 4). \quad (7)$$

For three distinct points, collinear and noncollinear triples give two different orbits; this is the extra degree-three term relative to the symmetric-group/Poisson pattern.

For nonzero vectors, ordered pairs consist of one independent orbit and one dependent orbit $y = ax$ for each $a \in \mathbb{F}_q^\times$, giving

$$[m_{(1,1)}] \mathcal{S} = q.$$

For an unordered pair, the dependent ratios other than 1 are identified by $a \leftrightarrow a^{-1}$. Together with the repeated and independent types, this gives

$$[m_{(2)}] \mathcal{S} = 2 + \left\lfloor \frac{q-1}{2} \right\rfloor.$$

3 Complete flags and failure of rank stability

For $S = G/B$, orbitals are the double cosets $B \backslash G/B$. Bruhat decomposition identifies these with the Weyl group, whence

$$[m_{(1,1)}] \mathcal{S}_{G,G/B} = |W|. \quad (8)$$

In type A_{n-1} this is $n!$, so even the degree-two coefficient cannot stabilize as rank grows.

4 Exact verification method

The accompanying marimo notebook performs the following independent steps.

1. Enumerate every finite-field matrix and retain the required determinant or invertibility condition.
2. Induce its permutation on nonzero vectors, normalized projective representatives, or incident point-hyperplane flags.
3. Count orbits directly on tuples of multisets, without using character or cycle formulas.
4. Extract the same coefficient from the actual cycle counts in (2).
5. Reconstruct every cycle count from fixed points of powers using (3).
6. At $(t_1, t_2) = (0.07, 0.11)$, compare the direct matrix-determinant average with the cycle-product average.

The last two numerical averages both equal 1.247556190416 for the $PSL_3(2)$ projective action, with difference below displayed precision.

Action	τ	direct orbits	cycle formula	claimed
$PSL_3(2)$ on $\mathbb{P}^2(\mathbb{F}_2)$	(1)	1	1	1
	(2)	2	2	2
	(1, 1)	2	2	2
	(3)	4	4	4
	(2, 1)	5	5	5
	(1, 1, 1)	6	6	6
$PSL_2(2)$ on $\mathbb{P}^1(\mathbb{F}_2)$	(3)	3	3	3
$GL_2(3)$ on nonzero vectors	(2)	3	3	3
	(1, 1)	3	3	3
$SL_2(3)$ on nonzero vectors	(1, 1)	4	4	4
$PGL_2(3)$ on projective points	(1, 1)	2	2	2
$PSL_2(3)$ on projective points	(1, 1)	2	2	2
$GL_2(2)$ on complete flags	(1, 1)	2	2	2
$GL_3(2)$ on complete flags	(1, 1)	6	6	6

The constructed orders are $|GL_3(2)| = 168$, $|GL_2(3)| = 48$, and $|SL_2(3)| = 24$; their projective images in degree two have orders 24 and 12. All displayed tests pass exactly.

Sharpness found by verification. The $SL_2(3)$ value 4 at $\tau = (1, 1)$ differs from the $GL_2(3)$ value 3. Here $n = |\tau| = 2$, so there is no unused complement on which to correct the determinant. Independent ordered pairs split into two $SL_2(3)$ orbits according to their determinant. This confirms that the proposed SL/GL agreement must retain the strict hypothesis $n > |\tau|$.

5 Scope of the conclusion

The matrix, orbit, fixed-point, and determinant routes agree for every representative case above. These computations verify the exact mechanism and the low-degree stable claims; the proofs of (1)–(8), rather than finite sampling, establish the formulas for all finite parameters in their stated ranges.